

Degree bound for Lasserre relaxation under regular sequence assumptions without infinity condition

ChatGPT (gpt-5.6-sol)*

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Abstract

We consider polynomial optimization in n variables with at least n equality constraints. We assume only that the full homogenizations of n selected equality polynomials form a homogeneous regular sequence. In particular, the homogenizing variable is allowed to be a zero-divisor modulo the resulting complete-intersection ideal, so the projective scheme may have components at infinity.

We introduce a saturation index τ that measures the nilpotent thickness of these components. If ρ is the least degree needed to represent every class in the affine quotient algebra, we prove an SOS-ideal certificate for the optimality gap of degree $M(\rho) + \tau$. We then establish the effective estimates

$$\rho \leq \sum_{i=1}^n (\deg g_i - 1), \quad \tau \leq \prod_{i=1}^n \deg g_i - \dim_{\mathbb{R}} \mathbb{R}[\mathbf{x}]/(g_1, \dots, g_n).$$

This gives an explicit finite-convergence bound for the SOS relaxation. Two examples show that the saturation penalty can be sharp and that τ need not be bounded by the usual complete-intersection regularity.

1 The open problem

This paper addresses an open problem posed by Zheng Hua, Zheng Qu in *Exactness and Effective Degree Bound of Lasserres Relaxation for Polynomial Optimization over Finite Variety* [1] (Mathematics of Operations Research, 2025), Section 8 (Conclusion and perspectives), page 21 (in the provided text, the paragraph beginning 'However, Assumption 1 is stronger ...'). The website catalogue entry is problem 19 (w4411879499_p0) of the [Open Problems in Operations Research](#) collection.

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Assume $k \geq n$ and $V_{\mathbb{C}}(g_1, \dots, g_k)$ is finite. Suppose the homogenizations $\bar{g}_1, \dots, \bar{g}_n \in \mathbb{R}[x_0, \dots, x_n]$ form a regular sequence (but do not assume the stronger condition that $g_1^\infty(z) = \dots = g_n^\infty(z) = 0$ has no solution in $\mathbb{C}^n \setminus \{0\}$).

Determine an explicit bound d (as a function of the degrees of $f, g_1, \dots, g_k, h_1, \dots, h_m$, and possibly other algebraic invariants naturally implied by the regular-sequence hypothesis) such that exactness at order d holds, i.e.

$$f - f^* \in I(g_1, \dots, g_k)_{2d} + QM(1, h_1, \dots, h_m)_{2d},$$

under the same type of nonsingularity/rank assumptions at minimizers used in the paper to ensure membership at some finite order.

Equivalently, establish effective degree bounds for exactness of Lasserres hierarchy when only the regular-sequence condition on $\bar{g}_1, \dots, \bar{g}_n$ is assumed, despite possible failure of the H -basis property.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Problem and main result

Let

$$R = \mathbb{R}[x_1, \dots, x_n]$$

and consider the polynomial optimization problem

$$f^* = \min_{\mathbf{x} \in \mathcal{F}} f(\mathbf{x}), \quad (2.1)$$

where

$$\mathcal{F} = \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) = 0 \quad (1 \leq i \leq k), \quad h_j(\mathbf{x}) \geq 0 \quad (1 \leq j \leq m)\}, \quad (2.2)$$

and $k \geq n$. We put $h_0 = 1$.

For a nonzero polynomial $p \in R$ of degree e , its full homogenization is

$$p^h(x_0, \mathbf{x}) = x_0^e p\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

Let

$$S = \mathbb{R}[x_0, x_1, \dots, x_n], \quad I_0 = (g_1, \dots, g_n) \subseteq R,$$

and define

$$J = (g_1^h, \dots, g_n^h) \subseteq S.$$

We shall assume that

$$g_1^h, \dots, g_n^h$$

is a homogeneous regular sequence in S .

It is important to distinguish J from the homogenization of the whole ideal I_0 . Throughout,

$$I_0^h := (p^h : p \in I_0). \quad (2.3)$$

The standard homogenization–saturation identity gives

$$K := I_0^h = J : x_0^\infty. \quad (2.4)$$

Define the saturation index

$$\tau := \min \{t \geq 0 : J : x_0^t = J : x_0^\infty\}. \quad (2.5)$$

Thus $\tau = 0$ precisely when J is already saturated with respect to x_0 .

Once R/I_0 is known to be finite-dimensional, put

$$r_0 := \dim_{\mathbb{R}}(R/I_0) \quad (2.6)$$

and define its interpolation degree by

$$\rho := \min \{q \geq 0 : R_{\leq q} \longrightarrow R/I_0 \text{ is surjective}\}. \quad (2.7)$$

For $1 \leq i \leq n$, set

$$d_i := \deg g_i, \quad D := \prod_{i=1}^n d_i, \quad N := \sum_{i=1}^n (d_i - 1).$$

For $q \geq 0$, define

$$M(q) := \max \left\{ \deg f, \max_{0 \leq j \leq m} (2q + \deg h_j), \max_{n < i \leq k} (q + \deg g_i) \right\}, \quad (2.8)$$

where the last maximum is omitted if $k = n$.

For an integer $L \geq 0$, let

$$I_L(g) := \left\{ \sum_{i=1}^k a_i g_i : \deg(a_i g_i) \leq L \text{ for every } i \right\}, \quad (2.9)$$

$$Q_L(h) := \left\{ \sum_{j=0}^m \sum_{\nu} s_{j\nu}^2 h_j : 2 \deg s_{j\nu} + \deg h_j \leq L \text{ for every } j, \nu \right\}. \quad (2.10)$$

We use the convention $\deg 0 = -\infty$.

Our main result is the following.

Theorem 2.1 (Effective certificate without the condition at infinity). *Assume that $\mathcal{F} \neq \emptyset$ and that*

$$g_1^h, \dots, g_n^h$$

is a homogeneous regular sequence in S . Then R/I_0 is finite-dimensional and $\mathcal{F}$ is finite. In particular, the minimum f^ in (2.1) exists.*

Suppose, in addition, that every global minimizer $\mathbf{x}^ \in \mathcal{F}$ satisfies*

$$\text{rank} [\nabla g_1(\mathbf{x}^*) \quad \cdots \quad \nabla g_k(\mathbf{x}^*)] = n. \quad (2.11)$$

Then

$$f - f^* \in I_{M(\rho)+\tau}(g) + Q_{M(\rho)}(h) \subseteq I_{M(\rho)+\tau}(g) + Q_{M(\rho)+\tau}(h). \quad (2.12)$$

Moreover,

$$\rho \leq N, \quad \tau \leq D - r_0, \quad (2.13)$$

and hence

$$f - f^* \in I_L(g) + Q_L(h), \quad L := M(N) + D - r_0. \quad (2.14)$$

Since feasibility implies $r_0 \geq 1$, the degree-only bound

$$L \leq M(N) + D - 1 \quad (2.15)$$

is also valid.

Define the SOS relaxation value

$$f_d^{\text{sos}} := \sup \{ \gamma \in \mathbb{R} : f - \gamma \in I_{2d}(g) + Q_{2d}(h) \}. \quad (2.16)$$

Corollary 2.2 (Finite convergence). *Under the hypotheses of Theorem 2.1,*

$$f_d^{\text{sos}} = f^*$$

for every integer d satisfying

$$2d \geq M(\rho) + \tau.$$

In particular, it is enough that

$$d \geq \left\lceil \frac{M(N) + D - r_0}{2} \right\rceil, \quad (2.17)$$

or, using only the degrees,

$$d \geq \left\lceil \frac{M(N) + D - 1}{2} \right\rceil.$$

Remark 2.3. *If x_0 is a non-zero-divisor on S/J , then*

$$J : x_0^\infty = J$$

and consequently $\tau = 0$. Thus the only new term caused by components at infinity is the additive saturation penalty τ .

3 Affine zero-dimensionality and saturation lifting

We begin with the algebraic consequences of the homogeneous regular sequence assumption.

Lemma 3.1 (Affine zero-dimensionality). *If $g_1^h, \dots, g_n^h$ is a regular sequence and $I_0 \neq R$, then*

$$\dim(R/I_0) = 0.$$

In particular, R/I_0 is a finite-dimensional $\mathbb{R}$ -algebra.

Proof. Because J is generated by a regular sequence of length n in the polynomial ring S of dimension $n + 1$,

$$\dim(S/J) = 1.$$

After inverting x_0 , dehomogenization gives

$$(S/J)_{x_0} \cong (R/I_0)[x_0, x_0^{-1}]. \quad (3.1)$$

The assumption $I_0 \neq R$ makes this localization nonzero. Therefore

$$\dim(R/I_0) + 1 = \dim((R/I_0)[x_0, x_0^{-1}]) = \dim(S/J)_{x_0} \leq 1.$$

It follows that $\dim(R/I_0) = 0$. A zero-dimensional finitely generated $\mathbb{R}$ -algebra is finite-dimensional over $\mathbb{R}$. $\square$

In our optimization problem, feasibility implies $I_0 \neq R$. Since $\mathcal{F}$ is contained in the real zero set of I_0 , Lemma 3.1 also shows that $\mathcal{F}$ is finite.

Lemma 3.2 (Homogenization and saturation). *With I_0 , J , and K as above,*

$$I_0^h = J : x_0^\infty.$$

In particular,

$$K : x_0 = K,$$

so multiplication by x_0 is injective on S/K .

Proof. Let $p \in I_0$, say

$$p = \sum_{i=1}^n a_i g_i.$$

Choose E at least as large as the degrees of p and all products $a_i g_i$. Homogenizing the identity to degree E gives

$$x_0^{E-\deg p} p^h = \sum_{i=1}^n x_0^{E-\deg(a_i g_i)} a_i^h g_i^h.$$

Thus some power of x_0 times p^h belongs to J , which proves

$$I_0^h \subseteq J : x_0^\infty.$$

Conversely, let F be homogeneous of degree E and suppose that $F \in J : x_0^\infty$. Then $F(1, \mathbf{x}) \in I_0$. If $e = \deg F(1, \mathbf{x})$, homogeneity gives

$$F = x_0^{E-e} (F(1, \mathbf{x}))^h.$$

The right-hand side belongs to I_0^h , proving the reverse inclusion. Finally, saturation implies

$$(J : x_0^\infty) : x_0 = J : x_0^\infty.$$

$\square$

The next lemma is the precise replacement for the usual H -basis argument.

Lemma 3.3 (Saturation lifting). *Let $r \in I_0$ have degree e . Then there exist $a_1, \dots, a_n \in R$ such that*

$$r = \sum_{i=1}^n a_i g_i \quad (3.2)$$

and

$$\deg(a_i g_i) \leq e + \tau \quad (1 \leq i \leq n). \quad (3.3)$$

Proof. By Lemma 3.2,

$$r^h \in K = J : x_0^\tau.$$

Consequently,

$$x_0^\tau r^h \in J.$$

The left-hand side is homogeneous of degree $e + \tau$. Taking the degree- $(e + \tau)$ homogeneous part of a representation in J , we obtain

$$x_0^\tau r^h = \sum_{i=1}^n A_i g_i^h,$$

where A_i is homogeneous of degree $e + \tau - d_i$, or is zero if that degree is negative. Setting $x_0 = 1$ yields

$$r = \sum_{i=1}^n A_i(1, \mathbf{x}) g_i,$$

and every product on the right has degree at most $e + \tau$. □

4 The interpolation-degree bound

We next prove that the complete-intersection regularity still controls the interpolation degree after saturation.

Proposition 4.1. *One has*

$$\rho \leq N = \sum_{i=1}^n (d_i - 1).$$

Proof. Put

$$B = S/J, \quad C = S/K.$$

The ring B is a one-dimensional homogeneous complete intersection. By Lemma 3.2, x_0 is a non-zero-divisor on C .

We claim that there is a real linear form $\ell \in S_1$ that is a non-zero-divisor on both B and C . Indeed, each ring has only finitely many associated primes. Since B is a one-dimensional Cohen–Macaulay ring, its associated primes are its minimal primes and do not include the irrelevant maximal ideal. Since x_0 is a non-zero-divisor on C , no associated prime of C contains x_0 . Hence no associated prime of either ring contains all of S_1 . Over the infinite field $\mathbb{R}$, finitely many proper linear subspaces cannot cover S_1 , so a common regular linear form exists.

The quotient map $B \twoheadrightarrow C$ induces a graded surjection

$$B/\ell B \twoheadrightarrow C/\ell C.$$

Because

$$g_1^h, \dots, g_n^h, \ell$$

is a regular sequence, the Hilbert series of $B/\ell B$ is

$$\text{Hilb}_{B/\ell B}(t) = \prod_{i=1}^n (1 + t + \dots + t^{d_i-1}). \quad (4.1)$$

The right-hand side has top degree N . Therefore

$$(C/\ell C)_q = 0 \quad (q > N).$$

Since ℓ is a non-zero-divisor on C , multiplication by ℓ gives exact sequences

$$0 \longrightarrow C_{q-1} \xrightarrow{\ell} C_q \longrightarrow (C/\ell C)_q \longrightarrow 0.$$

It follows that

$$\dim_{\mathbb{R}} C_q = \dim_{\mathbb{R}} C_{q-1} \quad (q > N).$$

Thus the Hilbert function of C is constant from degree N onward.

For every q , dehomogenization induces an isomorphism

$$C_q \cong \text{im}(R_{\leq q} \longrightarrow R/I_0). \quad (4.2)$$

Indeed, a polynomial of degree at most q may be homogenized to degree q , proving surjectivity onto the displayed image. If a homogeneous $F \in S_q$ dehomogenizes to an element of I_0 , then

$$F = x_0^{q-e} p^h$$

for some $p \in I_0$ of degree $e \leq q$; hence $F \in K$, proving injectivity.

The images in (4.2) are nested and exhaust the finite-dimensional algebra R/I_0 . Their dimensions are constant from degree N , so the degree- N image must already be all of R/I_0 . Therefore $\rho \leq N$. $\square$

5 The saturation index and the scheme at infinity

We now bound the saturation index by the total scheme-theoretic length at infinity.

Proposition 5.1. *With $D = \prod_{i=1}^n d_i$ and $r_0 = \dim_{\mathbb{R}}(R/I_0)$,*

$$\tau \leq D - r_0.$$

Proof. We first extend scalars from $\mathbb{R}$ to $\mathbb{C}$. Faithful flatness gives

$$(J : x_0^t) \otimes_{\mathbb{R}} \mathbb{C} = J_{\mathbb{C}} : x_0^t$$

for every t . It also preserves the stabilized saturation and detects equality of ideals. Hence the saturation index does not change after complexification. Moreover,

$$\dim_{\mathbb{C}}((R/I_0) \otimes_{\mathbb{R}} \mathbb{C}) = r_0.$$

Let

$$X = \text{Proj}(S_{\mathbb{C}}/J_{\mathbb{C}}).$$

Since $J_{\mathbb{C}}$ is a complete intersection of degrees $d_1, \dots, d_n$, X is a zero-dimensional projective scheme of total length

$$\text{length}(X) = D.$$

This also follows directly from

$$\text{Hilb}_{S_{\mathbb{C}}/J_{\mathbb{C}}}(t) = \frac{\prod_{i=1}^n (1 + t + \cdots + t^{d_i-1})}{1 - t}.$$

Let X_{∞} denote the set of points of X lying on $x_0 = 0$. For each $P \in X_{\infty}$, choose a linear form ℓ_P with $\ell_P(P) \neq 0$, and put

$$u_P := \frac{x_0}{\ell_P} \in \mathcal{O}_{X,P}.$$

Since u_P belongs to the maximal ideal of the local Artin algebra $\mathcal{O}_{X,P}$, it is nilpotent. Define

$$\nu_P := \min \{t \geq 1 : u_P^t = 0 \text{ in } \mathcal{O}_{X,P}\}.$$

We claim that

$$\tau = \max_{P \in X_{\infty}} \nu_P, \quad (5.1)$$

where the maximum over the empty set is defined to be zero.

Indeed, $S_{\mathbb{C}}/J_{\mathbb{C}}$ is Cohen–Macaulay of dimension one, hence $J_{\mathbb{C}}$ is unmixed and has no embedded components. Choose a homogeneous primary decomposition

$$J_{\mathbb{C}} = \bigcap_{P \in X} Q_P,$$

where Q_P is primary for the homogeneous prime corresponding to P .

If $P \notin X_{\infty}$, then $x_0 \notin \sqrt{Q_P}$, and hence

$$Q_P : x_0^t = Q_P \quad (t \geq 0).$$

If $P \in X_{\infty}$, the component disappears after coloning by x_0^t exactly when $u_P^t = 0$ in the corresponding local algebra. The use of $u_P = x_0/\ell_P$ causes no ambiguity because ℓ_P is a unit locally; primaryness also shows that local vanishing is equivalent to the required homogeneous membership.

Colons distribute over finite intersections. Therefore all affine primary components remain unchanged, while an infinity component disappears precisely when $t \geq \nu_P$. If an infinity component has not disappeared, localization at its prime shows that the colon ideal cannot yet equal $J_{\mathbb{C}} : x_0^{\infty}$. This proves (5.1).

Set

$$e_P := \text{length}_{\mathbb{C}} \mathcal{O}_{X,P}.$$

The chain

$$\mathcal{O}_{X,P} \supsetneq (u_P) \supsetneq \cdots \supsetneq (u_P^{\nu_P-1}) \supsetneq 0$$

is strict. Indeed, equality of two consecutive terms would imply $(u_P^j) = u_P(u_P^j)$, and Nakayama's lemma would force $(u_P^j) = 0$, contrary to the definition of ν_P . Consequently,

$$\nu_P \leq e_P.$$

The affine chart $x_0 \neq 0$ has coordinate algebra

$$\mathbb{C}[x_1, \dots, x_n]/I_{0,\mathbb{C}},$$

whose length is r_0 . Hence

$$D = r_0 + \sum_{P \in X_\infty} e_P.$$

Using (5.1),

$$\tau = \max_{P \in X_\infty} \nu_P \leq \max_{P \in X_\infty} e_P \leq \sum_{P \in X_\infty} e_P = D - r_0.$$

If X_∞ is empty, both τ and $D - r_0$ are zero. □

6 A quadratic-module identity in the equality quotient

Let

$$I = (g_1, \dots, g_k) \quad \text{and} \quad A = R/I.$$

Since $I_0 \subseteq I$ and R/I_0 is finite-dimensional, A is an Artinian $\mathbb{R}$ -algebra.

For $a \in A$, write

$$Q_A(h) := \left\{ \sum_{j=0}^m \sum_{\nu} b_{j\nu}^2 \overline{h_j} : b_{j\nu} \in A \right\},$$

where bars denote classes in A .

We first record two elementary facts about local Artin algebras.

Lemma 6.1 (Square roots and a lift of i). *Let B be a finite-dimensional local $\mathbb{R}$ -algebra with maximal ideal $\mathfrak{n}$.*

1. *If $B/\mathfrak{n} \cong \mathbb{R}$ and $b \in B$ has positive residue, then $b = v^2$ for some unit $v \in B$.*
2. *If $B/\mathfrak{n} \cong \mathbb{C}$, then there is an element $\iota \in B$ satisfying*

$$\iota^2 = -1.$$

Proof. The maximal ideal $\mathfrak{n}$ is nilpotent.

For the first assertion, the polynomial $T^2 - b$ has a simple root modulo $\mathfrak{n}$, namely the positive square root of the residue of b . A simple root lifts across a nilpotent ideal. Concretely, if v is a root modulo $\mathfrak{n}^r$, then $2v$ is a unit and the Newton correction

$$v' = v - \frac{v^2 - b}{2v}$$

is a root modulo $\mathfrak{n}^{2r}$. Since $\mathfrak{n}$ is nilpotent, finitely many corrections produce an exact square root.

For the second assertion, lift the element $i \in \mathbb{C} = B/\mathfrak{n}$ arbitrarily to B . Applying the same finite Newton procedure to $T^2 + 1$ produces an exact lift ι satisfying $\iota^2 = -1$. □

Proposition 6.2 (Local quadratic-module certificate). *Suppose every global minimizer satisfies the rank condition (2.11). Then*

$$\overline{f - f^*} \in Q_A(h). \tag{6.1}$$

Proof. The Artinian algebra A decomposes as a product of local Artin $\mathbb{R}$ -algebras. The residue field of each local factor is either $\mathbb{R}$ or $\mathbb{C}$. We construct the identity on every full local factor, including all of its nilpotent structure.

Let B first be a local factor with residue field $\mathbb{R}$. It corresponds to a real common zero a of $g_1, \dots, g_k$. Write

$$b := \overline{f - f^*} \in B.$$

If a is infeasible, there is some j for which $h_j(a) < 0$. By Lemma 6.1, there is a unit $v \in B$ such that

$$v^2 = -\overline{h_j}.$$

The identity

$$b = \left(\frac{b+1}{2}\right)^2 + \overline{h_j} \left(\frac{b-1}{2v}\right)^2 \quad (6.2)$$

then shows that $b \in Q_B(h)$.

Suppose next that a is feasible but not a global minimizer. Then

$$f(a) - f^* > 0.$$

Lemma 6.1 gives a square root of b in B . Thus b is an h_0 -square.

It remains to consider a global minimizer a . Let

$$\mathfrak{m}_a = (x_1 - a_1, \dots, x_n - a_n) \subseteq R$$

and let $\mathfrak{n}$ be the maximal ideal in the corresponding quotient-local factor B . Its cotangent space satisfies

$$\mathfrak{n}/\mathfrak{n}^2 \cong \mathfrak{m}_a / (\mathfrak{m}_a^2 + (g_1, \dots, g_k)). \quad (6.3)$$

The linear part of g_i at a is determined by $\nabla g_i(a)$. The rank condition using all equality gradients therefore implies

$$\mathfrak{n}/\mathfrak{n}^2 = 0.$$

Thus $\mathfrak{n} = \mathfrak{n}^2$, and Nakayama's lemma gives $\mathfrak{n} = 0$. Since the residue field is $\mathbb{R}$, it follows that $B = \mathbb{R}$. The residue of $f - f^*$ is zero at the minimizer, so $b = 0$ in the entire local factor.

Finally, suppose that B has residue field $\mathbb{C}$. Choose $\iota \in B$ with $\iota^2 = -1$ by Lemma 6.1. Every $b \in B$ is a sum of two squares:

$$b = \left(\frac{b+1}{2}\right)^2 + \left(\frac{\iota(b-1)}{2}\right)^2. \quad (6.4)$$

The Chinese remainder theorem identifies A with the product of these local factors. The componentwise identities glue by means of the corresponding idempotents. This produces an identity in A itself, not merely an identity on the set of closed points, and proves (6.1). $\square$

7 Degree bookkeeping and proof of the main theorem

We now lift Proposition 6.2 while retaining degree control.

Proof of Theorem 2.1. Lemma 3.1 proves that R/I_0 is finite-dimensional. The feasible set is therefore finite, so f^* exists.

Let

$$B_0 := R/I_0.$$

The natural map

$$B_0 \longrightarrow A = R/(g_1, \dots, g_k)$$

has kernel generated by the classes of

$$g_{n+1}, \dots, g_k.$$

By Proposition 6.2, there are finitely many elements $\alpha_{j\nu} \in A$ such that

$$\overline{f - f^*} = \sum_{j=0}^m \sum_{\nu} \alpha_{j\nu}^2 \overline{h_j}.$$

Lift the $\alpha_{j\nu}$ first to B_0 . The resulting difference belongs to the kernel of $B_0 \rightarrow A$, so there are elements $\beta_i \in B_0$, $n < i \leq k$, such that

$$\overline{f - f^*} = \sum_{j=0}^m \sum_{\nu} \alpha_{j\nu}^2 \overline{h_j} + \sum_{i=n+1}^k \beta_i \overline{g_i} \quad \text{in } B_0. \quad (7.1)$$

By the definition of ρ , every class in B_0 has a representative of degree at most ρ . We may therefore choose polynomials $s_{j\nu}, q_i \in R$ representing the classes in (7.1) such that

$$\deg s_{j\nu} \leq \rho, \quad \deg q_i \leq \rho.$$

Set

$$r := f - f^* - \sum_{j=0}^m \sum_{\nu} s_{j\nu}^2 h_j - \sum_{i=n+1}^k q_i g_i. \quad (7.2)$$

Equation (7.1) says precisely that $r \in I_0$.

The terms in (7.2) satisfy

$$\deg(s_{j\nu}^2 h_j) \leq 2\rho + \deg h_j$$

and

$$\deg(q_i g_i) \leq \rho + \deg g_i.$$

It follows from the definition of $M(\rho)$ that

$$\deg r \leq M(\rho). \quad (7.3)$$

Apply Lemma 3.3 to r . There are $a_1, \dots, a_n \in R$ such that

$$r = \sum_{i=1}^n a_i g_i$$

and

$$\deg(a_i g_i) \leq \deg r + \tau \leq M(\rho) + \tau.$$

Substituting this identity into (7.2) gives

$$f - f^* = \sum_{i=1}^n a_i g_i + \sum_{i=n+1}^k q_i g_i + \sum_{j=0}^m \sum_{\nu} s_{j\nu}^2 h_j.$$

The first n equality products have degree at most $M(\rho) + \tau$, while every other term has degree at most $M(\rho)$. This proves (2.12).

Propositions 4.1 and 5.1 give

$$\rho \leq N, \quad \tau \leq D - r_0.$$

Since $M(q)$ is nondecreasing in q ,

$$M(\rho) + \tau \leq M(N) + D - r_0.$$

This proves the explicit certificate (2.14). Finally, feasibility gives $r_0 \geq 1$, yielding (2.15). $\square$

Proof of Corollary 2.2. If $2d \geq M(\rho) + \tau$, Theorem 2.1 shows that $\gamma = f^*$ is feasible in the SOS relaxation (2.16). Hence

$$f_d^{\text{SOS}} \geq f^*.$$

Conversely, evaluation of any certificate

$$f - \gamma \in I_{2d}(g) + Q_{2d}(h)$$

at a feasible point shows that $f(\mathbf{x}) - \gamma \geq 0$. Therefore $\gamma \leq f^*$, and so

$$f_d^{\text{SOS}} \leq f^*.$$

Thus $f_d^{\text{SOS}} = f^*$. $\square$

8 Examples

The first example shows that the additive saturation penalty can be sharp.

Example 8.1 (Sharpness of $\tau \leq D - r_0$). *Let $q \geq 1$, let $R = \mathbb{R}[x, y]$, and set*

$$g_1 = x, \quad g_2 = x^{q+1} + y.$$

Using z as the homogenizing variable,

$$g_1^h = x, \quad g_2^h = x^{q+1} + z^q y.$$

Thus

$$J = (x, z^q y).$$

The two homogenizations form a regular sequence: modulo x , the second polynomial is the nonzero element $z^q y$ of the domain $\mathbb{R}[z, y]$.

For $0 \leq t \leq q$,

$$J : z^t = (x, z^{q-t} y),$$

and hence

$$K = J : z^\infty = (x, y), \quad \tau = q.$$

On the affine chart,

$$I_0 = (x, x^{q+1} + y) = (x, y),$$

so $r_0 = 1$. Since

$$D = 1 \cdot (q + 1) = q + 1,$$

we obtain

$$\tau = q = D - r_0.$$

The degree loss is also necessary for the saturation-lifting lemma. Indeed,

$$y = g_2 - x^q g_1$$

is a representation with product degree $q + 1$. On the other hand, a representation

$$y = a_1 g_1 + a_2 g_2$$

in which every product has degree strictly below $q + 1$ cannot use g_2 , because every nonzero multiple of g_2 has degree at least $q + 1$. The remaining generator $g_1 = x$ cannot generate y . Thus no smaller product-degree bound is possible.

The second example shows that the saturation index can exceed the complete-intersection regularity N .

Example 8.2 (The saturation index can exceed N). Let $R = \mathbb{R}[x, y]$ and

$$g_1 = xy - 1, \quad g_2 = x^2 - x.$$

With homogenizing variable z ,

$$J = (xy - z^2, x^2 - xz).$$

These two quadrics form a regular sequence. Their affine ideal is

$$I_0 = (x - 1, y - 1),$$

so

$$r_0 = 1.$$

The saturated homogeneous ideal is

$$K = J : z^\infty = (x - z, y - z).$$

Here

$$D = 2 \cdot 2 = 4, \quad N = (2 - 1) + (2 - 1) = 2.$$

The unique point at infinity is

$$[z : x : y] = [0 : 0 : 1].$$

On the chart $y = 1$, put

$$u = \frac{z}{y}, \quad v = \frac{x}{y}.$$

The two equations become

$$v - u^2 = 0, \quad v^2 - vu = 0.$$

After substituting $v = u^2$, the second equation becomes

$$u^4 - u^3 = u^3(u - 1).$$

In the local ring at $u = 0$, the factor $u - 1$ is a unit. Therefore the local algebra at infinity is

$$\mathcal{O}_{X,[0:0:1]} \cong \mathbb{C}[u]_{(u)}/(u^3).$$

The local parameter $u = z/y$ has nilpotence index 3. Consequently,

$$\tau = 3.$$

Thus

$$\tau = 3 > D - 2 = N = 2,$$

while simultaneously

$$\tau = 3 = D - r_0.$$

In particular, N alone does not bound the saturation index.

9 Concluding remarks

The regular-sequence hypothesis controls the complete-intersection Hilbert function but does not prevent projective components at infinity. Saturation separates the two effects:

- The interpolation degree of the affine quotient remains bounded by

$$\rho \leq N.$$

- The failure of the generator homogenizations to form an H -basis is measured exactly by the saturation index τ .
- The saturation index is controlled by the total scheme-theoretic length at infinity:

$$\tau \leq D - r_0.$$

Accordingly, removing the non-zero-divisor assumption on x_0 changes the certificate bound only through the additive term τ . The argument relies essentially on having n selected equalities in n variables whose full homogenizations form a regular sequence. It does not address the different regime with fewer than n equalities, sparsity-sensitive bounds, or possible refinements based on gradient ideals.

References

- [1] Zheng Hua, Zheng Qu. *Exactness and Effective Degree Bound of Lasserres Relaxation for Polynomial Optimization over Finite Variety*. Mathematics of Operations Research, 2025. <https://doi.org/10.1287/moor.2024.0483>